MLOps Pipeline — Heart Disease Prediction

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https://github.com/2025cs05025/ml_learning.git

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1. Introduction

1.1 Problem & dataset

Heart disease is one of the leading causes of death globally. Early detection and risk assessment from routine clinical measurements can improve outcomes and prioritisation. This project builds an end-to-end Machine Learning Operations (MLOps) pipeline to predict heart-disease risk from the UCI Heart Disease dataset, covering the full lifecycle: data acquisition, exploratory analysis, model development, experiment tracking with MLflow, automated unit tests, continuous integration and delivery, containerisation, optional Kubernetes deployment, and production-style monitoring with Prometheus and Grafana—aligned with common industry MLOps practice.

The UCI Heart Disease dataset is used (see data/). After preprocessing, records include 13 numeric input features and a binary target (0 = no disease, 1 = disease). Feature groups include demographics (age, sex), clinical measures (cp, trestbps, chol, fbs, restecg), exercise-related variables (thalach, exang, oldpeak, slope), and angiographic indicators (ca, thal).

Data source: UCI Machine Learning Repository (Heart Disease).

1.2 Technology stack

Component	Technology
Programming language	Python 3.9+ (CI / Docker image: 3.11)
ML framework	scikit-learn (pinned in requirements.txt)
API	FastAPI + Uvicorn (Flask in requirements is optional; main API is FastAPI)
Experiment tracking	MLflow (local ./mlruns)
Testing	pytest (tests/test_pipeline.py)
CI/CD	GitHub Actions (.github/workflows/ci.yml)
Containerisation	Docker + Docker Compose
Orchestration	Kubernetes manifests under k8s/ (e.g. Minikube)
Monitoring	Prometheus + Grafana (monitoring/)

1.3 Official repository

Item	Value
URL	https://github.com/2025cs05025/ml_learning.git
Clone	git clone https://github.com/2025cs05025/ml_learning.git
Root folder	ml_learning
Author / ID	SANDIP BHATTACHARYYA — 2025cs05025

Work dir	Repo root so data/, src/, models/ match README

1.4 Deliverables map (repo paths)

Required element	Location in this repository
Application & library code	src/ (api/, config/, data_preprocessing/, eda/, model_training/, …)
Dockerfile(s) & Compose	Dockerfile at repo root; docker-compose.yml for API + Prometheus + Grafana
Python dependencies	requirements.txt (pinned scikit-learn for pickle compatibility)
Conda environment file	Not in repo. Use conda + pip install -r requirements.txt, or add env.yml if required
Raw & cleaned datasets	data/heart_disease_UCI_dataset.csv (bundled raw); data/heart_disease_processed_dataset.csv (generated by EDA preprocess)
Dataset download / provenance	README has the UCI link. Raw CSV is already in data/ (no download script)
EDA, training, inference	Python modules: src/eda/eda.py, src/model_training/train.py, src/model_training/inference.py (no .ipynb in-tree; notebooks optional)
Unit / integration tests	tests/ (pytest; pytest.ini sets testpaths=tests, pythonpath=src)
CI/CD workflow	.github/workflows/ci.yml (GitHub Actions; no Jenkinsfile)
Deployment manifests	k8s/ (Kustomize YAML). No Helm chart; you can add one if the course asks
Reporting screenshots	screenshots/*.png (embedded in §12.1 when this report is regenerated)
Monitoring config	monitoring/prometheus.yml; monitoring/grafana/ (dashboard JSON + provisioning)
Generated model artifacts (local/CI)	models/best_model.pkl, models/feature_names.pkl, models/training_metadata.json (recreated by train.py)
Experiment tracking store (local)	./mlruns/ for MLflow (often gitignored; CI can upload a zip artifact)

2. Setup & Installation Instructions

This section gives a complete installation narrative: what must exist on the machine before you start, how to create an isolated Python environment, how dependencies are pinned for reproducible unpickling of the model, and which optional tools unlock Docker Compose and Kubernetes demos. For edge cases (port conflicts, Minikube drivers, Apple Silicon), the README troubleshooting section is the authoritative extension of these steps.

2.1 Prerequisites

- Python 3.9 or newer on the host (3.11 is used in CI and in the API Docker image). Verify with: python3 --version.
- Git, to clone and push the repository.
- A C compiler toolchain may be required on some platforms when pip builds binary wheels (gcc is also installed in the Dockerfile).
- Docker Desktop or an equivalent OCI runtime (optional) for docker build, docker compose, and image load into Minikube.
- kubectl and Minikube (optional) only if you demonstrate the k8s/ manifests.

2.2 Local setup

Create a dedicated virtual environment so that scikit-learn, MLflow, FastAPI, and pytest versions match those used during training and in CI. The project pins scikit-learn==1.6.1 because joblib pickles are not guaranteed compatible across arbitrary sklearn major/minor versions. After activation, upgrade pip, then install from requirements.txt. Full detail (ports, troubleshooting, K8s): README.md in the repository.

Step	Command / action	Notes
1	git clone https://github.com/2025cs05025/ml_learning.git	Then: cd ml_learning
2	python3 -m venv .venv	Activate: source .venv/bin/activate (Windows: .venv\Scripts\activate)
3	pip install -r requirements.txt	Pins scikit-learn for pickle compatibility
4	python3 src/eda/eda.py all	Writes cleaned CSV + screenshots/
5	python3 src/model_training/train.py	GridSearch + MLflow; writes models/ and mlruns/
6	python3 -m pytest tests/ -v	Unit tests
7	python3 src/api/api.py	Or: uvicorn api.api:app --app-dir src --host 0.0.0.0 --port 8000
8	python3 -m mlflow ui --backend-store-uri ./mlruns --host 127.0.0.1 -p 5050	Experiment UI

Shell sequence (repository root):

```
git clone https://github.com/2025cs05025/ml_learning.git
cd ml_learning
python3 -m venv .venv
python3 -m pip install --upgrade pip
pip install -r requirements.txt
python3 src/eda/eda.py all
python3 src/model_training/train.py
python3 -m pytest tests/ -v
```

```
python3 src/api/api.py
# Swagger UI: http://127.0.0.1:8000/docs
```

Verification	Expected
After pytest	Imports and paths OK
After train	models/best_model.pkl + mlruns/

3. Data Acquisition & Exploratory Data Analysis

Step	What we did	Output / path
Source	UCI Heart Disease CSV in repo	data/heart_disease_UCI_dataset.csv
Clean	pre_processing_data.py (missing tokens, types, binary target)	In-memory matrix; same code in train/API
EDA CLI	src/eda/eda.py preprocess | eda | all	data/heart_disease_processed_dataset.csv
Plots	Histograms, correlation, class balance, age×target	screenshots/*.png (default names)
Re-run	Single command for full EDA + preprocess	python3 src/eda/eda.py all

3.1 Exploratory Data Analysis

EDA — Key Insights:

- Age distribution: ranges from 29 to 77 years with a mean of ~54 years; slightly right-skewed.
- Gender: majority male (~68.3% male, ~31.7% female).
- Chest pain: type 0 is the most common (~47.2% of cases).
- Blood pressure: mostly between 120–140 mmHg.
- Fasting blood sugar: most patients below 120 mg/dl (85.1%).
- Heart disease prevalence: slightly more than half have heart disease (~54.5%).
- Correlations: age, exercise-induced angina, ST depression (oldpeak), and maximum heart rate show strong associations with the target.

4. Feature Engineering & Model Development

Model	Pipeline (sklearn)
Logistic regression	SimpleImputer(median) → StandardScaler → LogisticRegression(liblinear, class_weight=balanced, max_iter=2000)
Random forest	SimpleImputer(median) → RandomForestClassifier(class_weight=balanced); no scaler

4.1 Data splits

Stage / split	Approx. fraction	Rows used for	Implementation notes
Full cleaned dataset	100%	EDA plots, optional exploration	heart_disease_processed_dataset.csv
Training split	~80%	GridSearchCV, refit, CV metrics, MLflow train metrics	stratify=y, random_state=42
Hold-out test	~20%	Final metrics, ROC/CM plots, model selection vs other family	Never inside CV folds
CV folds (×5)	Training only	Hyperparameter scoring (ROC-AUC)	StratifiedKFold, shuffle, random_state=42
Inference / API	N/A	New patients or batch CSV	No split; apply fitted Pipeline from best_model.pkl

4.2 Hyperparameter search & evaluation

Setting	Value
Tuning	GridSearchCV + StratifiedKFold(5), shuffle, random_state=42
Scoring	roc_auc; refit on full train split
Winner	Highest hold-out test ROC-AUC →

	models/best_model.pkl

Parameter	Grid
lr__C	0.01, 0.1, 1.0, 10.0
rf__n_estimators	100, 200
rf__max_depth	None, 6, 12
rf__min_samples_leaf	1, 2

Model	Test metrics
Logistic Regression	roc_auc=0.9015, accuracy=0.8525, f1=0.8302
Random Forest	roc_auc=0.8755, accuracy=0.8033, f1=0.7778

Training metadata	Value
Best model (metadata)	Logistic Regression
saved_at_utc (metadata)	2026-05-06T15:05:15.015361+00:00
CV / random_state	cv_splits=5, random_state=42
sklearn (metadata)	1.6.1

Plot artifact	Path
CM — logistic regression	screenshots/cm_logistic_regression.png
CM — random forest	screenshots/cm_random_forest.png
ROC — logistic regression	screenshots/roc_logistic_regression.png
ROC — random forest	screenshots/roc_random_forest.png

5. MLflow

Topic	Detail	Notes
Code	src/model_training/train.py	Logs each family after GridSearchCV
Store	./mlruns	./mlruns (local file store)
Experiment	heart_disease_prediction	UI lists runs
Logged	params, CV + test metrics, plot	Optional sklearn model log

	files	
JSON summary	models/training_metadata.json	Best model + metrics
UI tabs	Parameters / Metrics / Artifacts	Test ROC-AUC picks winner in code

MLflow UI command:

```
python3 -m mlflow ui --backend-store-uri ./mlruns --host 127.0.0.1 -p 5050
```

6. Artifacts & Batch Scoring

File	Role	Used by
models/best_model.pkl	Fitted Pipeline	inference.py, api.py
models/feature_names.pkl	Column order	api.py
models/training_metadata.json	Metrics / versions	Report, audit

Reproducibility	How
Versions	requirements.txt sklearn pin
Splits	random_state=42 in train & CV
Cleaning	Shared pre_processing_data.py

Batch inference command:

```
python3 src/model_training/inference.py --output data/batch_predictions.csv
```

7. CI/CD Pipeline & Automated Testing

7.1 Automated tests

pytest configuration:

pytest.ini sets pythonpath=src and testpaths=tests. Tests under tests/ in tests/test_pipeline.py (data load/clean, split checks, both sklearn pipelines, predict_proba, PatientData schema).

Code block — run tests locally:

```
python3 -m pytest tests/ -v --tb=short
```

CI job	Purpose
flake8	Lint src/, tests/
pytest	JUnit + log artifacts
train	EDA + train.py → artifact zip
docker smoke	Build image; /health, /predict

ci.yml excerpt:

```
# .github/workflows/ci.yml — CI/CD for MLOps assignment (course S2-
25_AMLCSZG523).
# Author: SANDIP BHATTACHARYYA — BITS Pilani ID 2025cs05025
#
# ── When does this run?
──────────────────────────────────────────────────────────
#   • push / merge to branches: main, master only (other branches: open a PR)
#   • pull_request targeting main or master
#   • workflow_dispatch (Actions tab → CI/CD Pipeline → Run workflow)
#
# ── Four jobs (sequential)
──────────────────────────────────────────────────────
#   1. lint — flake8
#   2. testing — pytest + upload test-results (pytest-results.xml)
#   3. training-model — preprocess → train.py → upload ml-training (models/ +
mlruns/)
#   4. docker-build-smoke — needs training-model; docker build + /health +
/predict smoke
#
#   Later jobs show "Queued" until the previous job finishes. That is normal.
#
# ── How long? (ballpark)
──────────────────────────────────────────────────────────
#   Often ~15–35 minutes total. Slow parts: pip install (per job),
GridSearchCV,
#   Docker build. training-model: 45m; lint/testing: 15m each.
#
name: CI/CD Pipeline

on:
  push:
    branches: [main, master]
  pull_request:
    branches: [main, master]
  workflow_dispatch:
```

Screenshot (save to screenshots/)	Shows
Actions list	Green workflow run
Job graph	lint → test → train → docker
Logs	pytest tail, train upload, docker curl
Artifacts	XML, training zip, smoke logs
Compose / k8s	Services or pods up

8. Container (API)

Image	Detail
Base	python:3.11-slim
App	PYTHONPATH=/app/src; Uvicorn api.api:app
Health	GET /health

Dockerfile:

```
# Dockerfile — production-style image for the FastAPI inference API
(assignment Task 6).
# Author: SANDIP BHATTACHARYYA — BITS Pilani ID 2025cs05025
#
# Expects trained artifacts under models/ at build time (best_model.pkl,
feature_names.pkl).
# Build:  docker build -t heart-disease-api:latest .
# Run:    docker run --rm -p 8000:8000 heart-disease-api:latest
# Stack:  prefer docker-compose.yml for API + Prometheus + Grafana.
#
# ── Stage 1: base image
─────────────────────────────────────────────────────────

FROM python:3.11-slim

WORKDIR /app

# PYTHONPATH so ``api.api`` resolves to ``src/api/api.py``
ENV PYTHONDONTWRITEBYTECODE=1 \
    PYTHONUNBUFFERED=1 \
    PYTHONPATH=/app/src \
    MODEL_PATH=/app/models/best_model.pkl \
    FEATURE_PATH=/app/models/feature_names.pkl \
    API_LOG_PATH=/app/logs/api_requests.log \
    PIP_ROOT_USER_ACTION=ignore

RUN apt-get update && apt-get install -y --no-install-recommends \
    gcc \
```

```
    curl \
    && rm -rf /var/lib/apt/lists/* \
    && mkdir -p /app/logs

COPY requirements.txt .
RUN pip install --no-cache-dir --upgrade pip \
    && pip install --no-cache-dir -r requirements.txt

COPY src/ ./src/
COPY models/ ./models/

EXPOSE 8000

HEALTHCHECK --interval=30s --timeout=10s --start-period=20s --retries=3 \
    CMD curl -fsS http://127.0.0.1:8000/health >/dev/null || exit 1
```

Build & run (host shell):

```
docker build -t heart-disease-api:latest .
docker run --rm -p 8000:8000 heart-disease-api:latest
```

8.1 API surface (FastAPI)

Endpoint	Method	Description
/health	GET	Liveness; pipeline loaded
/predict	POST	JSON body → prediction, label, confidence, risk_level
/metrics	GET	Prometheus text format
/docs	GET	Swagger UI

Smoke checks (API on port 8000):

```
curl -sS http://127.0.0.1:8000/health
curl -sS -X POST http://127.0.0.1:8000/predict \
  -H "Content-Type: application/json" \
  -d
'{"age":55,"sex":1,"cp":2,"trestbps":130,"chol":250,"fbs":0,"restecg":0,"thal
ach":150,"exang":0,"oldpeak":1.0,"slope":1,"ca":0,"thal":2}'
```

9. Deployment

Compose service	URL	Role
API	http://127.0.0.1:8000	FastAPI
Prometheus	http://127.0.0.1:9090	Scrapes /metrics

Grafana	http://127.0.0.1:3000	Dashboards (README: creds)

Command / path	Purpose
docker compose up -d --build	Stack: API + Prometheus + Grafana. Refer README for more details https://github.com/2025cs05025/ml_learning#8-docker-api-image-only
kubectl apply -k k8s/	Optional K8s (Kustomize); See README for Minikube image load, namespaces, and port-forwards. refer: https://github.com/2025cs05025/ml_learning#kubernetes-deployment-minikube

10. Monitoring

Piece	Location	Role
Logs	src/api/api.py; API_LOG_PATH → logs/	Structured request log
Metrics	GET /metrics on API	Counters, latency, predictions (prometheus_client)
Dashboard	monitoring/grafana/heart_disease_api.json	Grafana panels
Scrape config	monitoring/prometheus.yml	Compose scrape target

prometheus.yml excerpt:

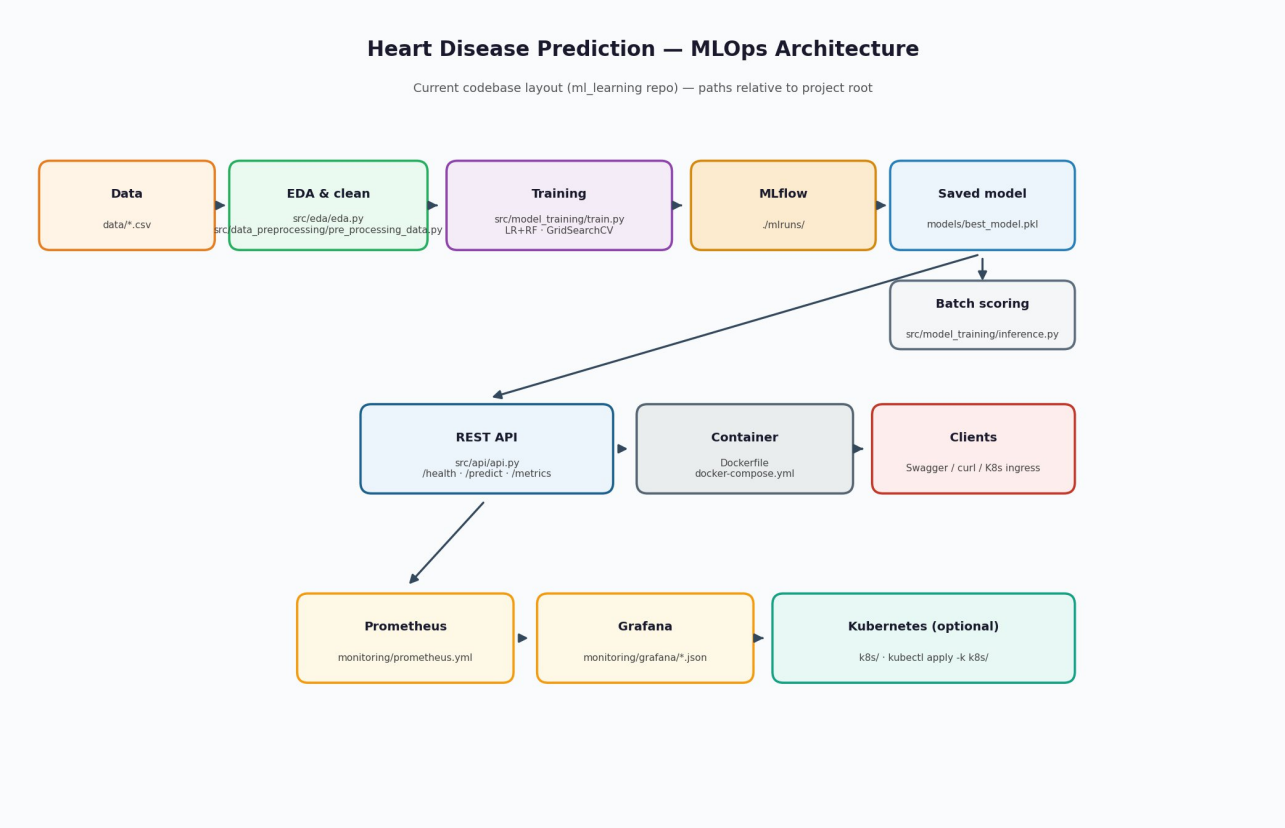
```
# Prometheus scrape config for Docker Compose (service name heart-disease-
api).
# Mounted into the prometheus container via docker-compose.yml.
# Author: SANDIP BHATTACHARYYA — BITS Pilani ID 2025cs05025

global:
  scrape_interval: 15s
  evaluation_interval: 15s

scrape_configs:
  # Faster scrape during local Compose tests so counters appear in TSDB
quickly after /predict.
  - job_name: heart-disease-api
    scrape_interval: 5s
    # Prometheus requires scrape_timeout <= scrape_interval.
    scrape_timeout: 4s
```

```
    metrics_path: /metrics
    static_configs:
      - targets:
          - heart-disease-api:8000
```

11. Architecture



12. Evidence reference

Please refer:

1. Screenshots present in https://github.com/2025cs05025/ml_learning/tree/main/screenshots
2. https://github.com/2025cs05025/ml_learning/blob/main/reports/Heart_Disease_MLOps_Evidence_Sandip_Bhattacharyya_2025cs05025.pdf